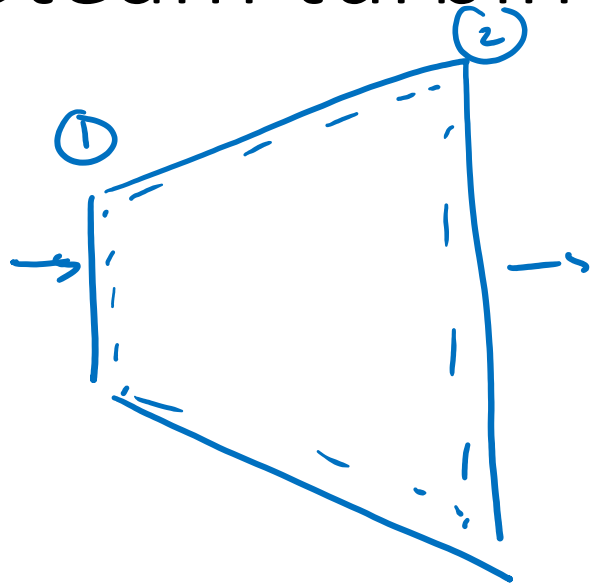


An Emirates airplane is shown from a low angle, flying upwards and to the right. It is leaving four distinct, thick white smoke trails that fan out behind it against a clear blue sky. The aircraft is white with a red and black livery, and the word 'Emirates' is visible on the side of the fuselage.

ME 300:
Engineering Thermodynamics

Example:
Steady Flow
Energy Equation

Steam turbine



Assume

- Steady-flow
 - S.C.S.
 - $\Delta ke = \Delta pe = 0$
- $\Rightarrow \dot{Q} = 0$

$$\dot{m} = 38.7 \text{ kg/s}$$

$$T_1 = 780 \text{ K}$$

$$P_1 = 10 \text{ MPa}$$

$$T_2 = 320 \text{ K}$$

$$x_2 = 0.88$$

$\Rightarrow$ produce work

$$\dot{m}(\Delta h + \cancel{\Delta ke} + \cancel{\Delta pe}) = \cancel{\dot{Q}} - \dot{W}$$

$$\dot{m} \Delta h = -\dot{W}$$

$$\dot{W} = -\dot{m} \Delta h$$

↑
 $h_1 \quad h_2$



What is the
purpose of a
turbine?

Steam turbine

State 1: 780 K
10 MPa

$\Rightarrow$ D.2 @ 10 MPa $T > T_{\text{sat}} \rightarrow$ vapor

$\Rightarrow$ D.3 @ 10 MPa $\Rightarrow$ $h_1 = 3392.8$ kJ/kg

Steam turbine

State 2 : $T_2 = 320$ K $x_2 = 0.88 \Rightarrow$ mixture

$$\Rightarrow \text{D.1} \rightarrow h_2 = x_2 h_g + (1-x_2) h_f$$

$$= (0.88)(2585.7) + (0.12)(196.17)$$

$$h_2 = 2298.96 \text{ kJ/kg}$$

$$\dot{W} = -\dot{m}(h_2 - h_1) = -(38.7)(2298.96 - 3392.8)$$

$$\dot{W} = 42331.6 \frac{\text{kJ}}{\text{s}} \cdot \text{kW}$$